

# Tethered 6U CubeSat

Tether range, relative attitude, state of the art and 3D imaging

CUBICS 2026 · Stream 1 · Team meeting, October 2026

One tether, two 3U halves of 4 kg, circular orbit at 550 km.

## Four answers for today

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|                          |                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Tether range</b>      | Vertical only. 500 m to 1 km is feasible: 3.6 to 7.2 mN of tension, about 3.5 h to deploy.                                    |
| <b>Relative attitude</b> | Exploit the mechanics: the tether and its anchor joint give the direction between the halves, without relying on electronics. |
| <b>State of the art</b>  | Tethers have flown since 1966. Most failures happen during deployment.                                                        |
| <b>3D photos</b>         | Not with a 0.5 to 1 km tether baseline: height errors near 1 km. Yes with fore/aft camera pointing: about 3.5 m.              |
| <b>Lifetime</b>          | Reel the tether in when not in use: about 6 years instead of 1.6 years at 550 km.                                             |

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# The tether is held taut by the gravity gradient

Circular orbit of radius  $a = R_{\oplus} + h$ :

$$n = \sqrt{\frac{\mu}{a^3}} = 1.0948 \times 10^{-3} \text{ rad/s}, \quad P = \frac{2\pi}{n} = 95.6 \text{ min.}$$

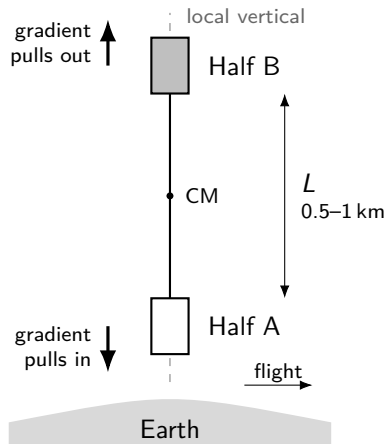
A body forced to turn at rate  $n$  at radius  $a + l$  lacks

$$n^2(a + l) - \frac{\mu}{(a + l)^2} \simeq 3 n^2 l$$

of inward acceleration. The tether supplies it:

$$T = 3 n^2 m_{\text{eq}} L, \quad m_{\text{eq}} = \frac{m_1 m_2}{m_1 + m_2} = 2 \text{ kg.}$$

$$T = 3.6 \text{ mN (500 m)} \quad T = 7.2 \text{ mN (1 km)}$$



# Only the vertical direction separates the two halves

Relative motion in the orbiting frame ( $x$  radial,  $y$  along track,  $z$  out of plane):

$$\ddot{x} - 2n\dot{y} - 3n^2x = f_x, \quad \ddot{y} + 2n\dot{x} = f_y, \quad \ddot{z} + n^2z = f_z.$$

| Direction            | Free motion                        | Tether |
|----------------------|------------------------------------|--------|
| Radial ( $x$ )       | $+3n^2x$ : the halves move apart   | taut   |
| Along track ( $y$ )  | no restoring or separating term    | slack  |
| Out of plane ( $z$ ) | $-n^2z$ : the halves move together | slack  |

A tilt  $\varphi$  from the vertical swings back (constant  $L$ ):

$$\ddot{\varphi} + 3n^2 \sin \varphi \cos \varphi = 0 \Rightarrow \omega_{\text{lib}} = \sqrt{3} n, \quad 2\pi/\omega_{\text{lib}} = 55.2 \text{ min.}$$

**A diagonal tether is a swing, not a configuration.** Holding  $\varphi = 30^\circ$  at 1 km needs  $3n^2 m_{\text{eq}} L^2 \sin \varphi \cos \varphi = 3.1 \text{ N m}$ , about 3000 times a CubeSat reaction wheel.

## 500 m to 1 km with one tether: the key numbers

| Quantity                                  | Expression             | $L = 500 \text{ m}$      | $L = 1 \text{ km}$       |
|-------------------------------------------|------------------------|--------------------------|--------------------------|
| Tension                                   | $3n^2 m_{\text{eq}} L$ | 3.6 mN                   | 7.2 mN                   |
| Elastic stretch                           | $TL/(EA)$              | 0.6 mm                   | 2.3 mm                   |
| Tether mass, $\varnothing 0.2 \text{ mm}$ | $\rho_c AL$            | 22 g                     | 44 g                     |
| Tether frontal area                       | $d L$                  | $0.10 \text{ m}^2$       | $0.20 \text{ m}^2$       |
| Controlled deployment, $\lambda = n/2$    | $\ln(L/L_0)/\lambda$   | 3.2 h                    | 3.5 h                    |
| Controlled retrieval, $\lambda = n/5$     | $\ln(L/L_0)/\lambda$   | 7.9 h                    | 8.8 h                    |
| Orbit life at 550 km, deployed            | drag model             | $\approx 2.5 \text{ yr}$ | $\approx 1.6 \text{ yr}$ |

$EA = 3142 \text{ N}$ ,  $L_0 = 1 \text{ m}$ , mean solar activity. Stowed:  $\approx 6 \text{ yr}$ .

## Once deployed, the length stays fixed

The winch sets  $L$ . More tension does not lengthen the tether; it only pulls it tighter.

### Elastic stretch

$$\Delta L_{\text{el}} = \frac{T L}{EA}$$

$$\leq \mathbf{2.3 \text{ mm}}$$

at 1 km and full tension

### Natural swing

$$2\pi/(\sqrt{3} n)$$

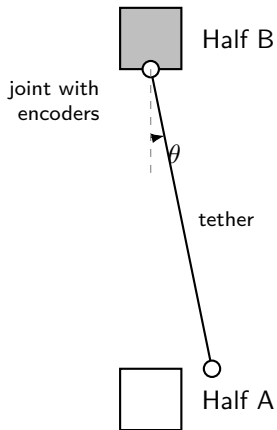
**55 min**

period around the vertical

The tension is fixed by the length:  $T = 3n^2 m_{\text{eq}} L$ . Paying out faster only adds swing and braking load.

# Relative attitude: exploit the mechanics

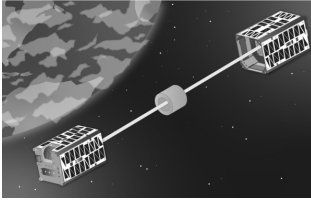
- The tether is a physical link: it already points from one half to the other.
- A passive joint at each anchor, with angle encoders, reads that direction directly.
- The measurement is mechanical: it does not drift and does not depend on extra electronics.
- The joined state gives the mechanical zero for calibration.



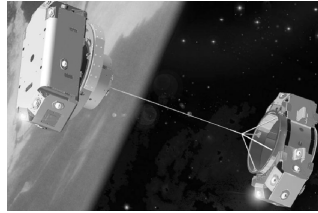
## Tether heritage: deployment is the main risk

| Mission          | Year | Length         | Result                                               |
|------------------|------|----------------|------------------------------------------------------|
| Gemini XI        | 1966 | 30 m           | First tether between two spacecraft                  |
| SEDS-1           | 1993 | 20 km          | Full deployment; end mass deorbited                  |
| SEDS-2           | 1994 | 19.7 km        | Closed-loop braking law validated                    |
| TSS-1R           | 1996 | 19.7 km        | <b>Tether broke by an electrical arc</b>             |
| TiPS             | 1996 | 4 km           | Libration damped from 40° to 7.5°                    |
| ATE <sub>x</sub> | 1999 | 22 m of 6.2 km | <b>Tether went slack; deployment stopped</b>         |
| YES2             | 2007 | 31.7 km        | Longest structure deployed in orbit                  |
| TEPCE            | 2019 | 1 km           | Two CubeSat halves split by a stacer spring at 4 m/s |

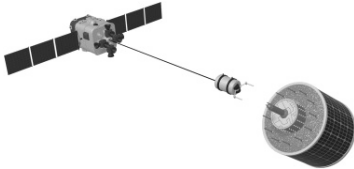
# Previous tethered systems



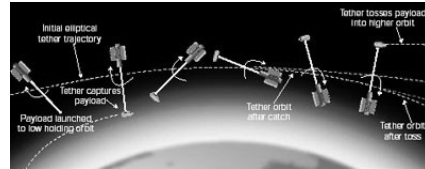
Two CubeSats joined by a tether



Two end bodies joined by a tether



Tether between a spacecraft and a target



Momentum exchange: catch and toss

# Where our proposal fits

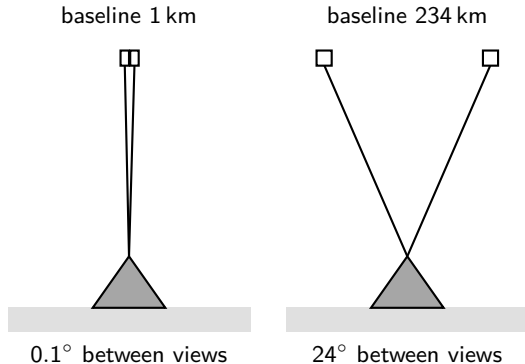
## Already demonstrated

- Splitting a CubeSat in two with a spring (TEPCE, 2019)
- Kilometre-class deployment with braking (SEDS-2, TiPS)
- Gravity-gradient stabilisation along the vertical
- Libration damping with the winch

## New in this proposal

- One tether with a winch that both deploys and reels in
- Tether direction measured at the anchor
- Tether reeled in between campaigns to extend the mission
- Earth imaging from both halves

## 3D needs two very different viewing angles



A point of height  $h$  shifts between the photos by

$$p = h \frac{B}{H},$$

so the height error is

$$\sigma_h = \frac{\sigma_p}{B/H}.$$

$B$  baseline,  $H = 550$  km,  $\sigma_p$  matching error (0.3 pixel = 1.5 m with 5 m pixels).

## A 0.5 to 1 km baseline gives height errors near 1 km

Tether baseline, even if horizontal:  $B/H = L/H$ , with  $H = 550$  km and 5 m pixels.

$L = 1$  km

$\sigma_h \approx 825$  m  
height error

$L = 500$  m

$\sigma_h \approx 1.65$  km  
height error

1 km-high mountain

**0.36 pixel**  
shift between the two photos

A vertical tether gives no stereo at all: both cameras look along the same line.

## What works: pointing the cameras

| Option                               | $B/H$                | $\sigma_h$ (5 m / 20 m pixels) | Notes                  |
|--------------------------------------|----------------------|--------------------------------|------------------------|
| Fixed, A $+12^\circ$ / B $-12^\circ$ | 0.43                 | 3.5 m / 14 m                   | Sites under the track  |
| Gimbal $\pm 30^\circ$                | 0.40                 | 3.8 m / 15 m                   | Chosen sites, 2–4 days |
| Tether baseline, 1 km                | $1.8 \times 10^{-3}$ | 825 m / 3.3 km                 | <b>Not usable</b>      |

The 3D comes from the satellite moving 234 km along its orbit in 31 s, not from the tether.

# Risks

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|                                 |                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Retrieval</b>                | Reeling in amplifies the swing ( $-2\dot{L}/L > 0$ ). It needs libration control and a slow rate: 7.9 to 8.8 h.         |
| <b>Lifetime</b>                 | Reel the tether in when not in use: about 6 yr instead of 1.6 yr at 550 km. Deploy it at end of life to reenter faster. |
| <b>Single tether</b>            | One cut by a small debris particle ends the mission. Use a braided or multi-strand line.                                |
| <b>Debris and admissibility</b> | A 1 km object has a larger collision cross-section. Confirm with the CSA.                                               |
| <b>Model accuracy</b>           | Orbit lives come from a simple drag model; a formal STELA or DRAMA study is needed.                                     |

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# Decisions for the team

- 1 **Tether length:** 500 m or 1 km, with the tether stowed between campaigns.
- 2 **Relative attitude:** joint with encoders at each anchor as the main measurement.
- 3 **Imaging claim:** drop tether stereo; use fore/aft or gimbal cameras for 3D.
- 4 **Questions to the CSA by 9 November:** launch altitude, tether admissibility, deorbit case.
- 5 **Analysis:** formal orbit-life study with STELA or DRAMA.